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**SIMATS ENGINEERING
ASSESSMENT TOOL 2**

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO4:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2,BL6)

Assessment Tool Weightage: 30%

13.6 48

Annexure A

Industry Problem-Solving Task

1. A star topology is ideal for a 20-person startup because all devices connect to a central switch, making the network easy to manage and troubleshoot. At the core sits a Layer 2/Layer 3 managed switch (such as a Cisco Catalyst or Netgear ProSafe) with at least 24 ports to accommodate all workstations, printers, and VoIP phones with room to grow. Each device connects to the switch using Cat6 UTP cables, which support Gigabit speeds up to 100 meters — more than sufficient for any typical office floor plan. The switch uplinks to a router/firewall (such as a Cisco RV series or pfSense appliance) that handles internet access, DHCP, and network address translation. A Wi-Fi access point connected to the switch extends wireless coverage for laptops and mobile devices.

3. For a small classroom lab with a tight budget, bus topology offers the lowest hardware cost because all computers share a single common communication line — the bus — eliminating the need for a central switch or hub. A single coaxial cable (RG-58, 50-ohm) or, in a modern variation, a single Cat5e/Cat6 cable daisy-chained through an inexpensive unmanaged hub runs along the length of the room. Each computer connects to the backbone using a T-connector and a BNC connector in a classic coaxial setup, with terminators (50-ohm) placed at both ends of the cable to absorb signals and prevent reflection. For a more contemporary low-cost implementation, a single inexpensive 8- or 16-port unmanaged switch replaces the terminators and uses Cat5e UTP patch cables to each workstation — achieving bus-like simplicity at minimal cost. The primary limitation is that the network is susceptible to collisions and a single cable break disrupts all connected machines, but for a supervised classroom environment with light usage (internet browsing, file sharing), this trade-off is entirely acceptable. No dedicated server is required; a teacher's PC can share resources using built-in Windows/Linux file sharing.

4. A large university campus serves thousands of users across multiple buildings — lecture halls, labs, libraries, dormitories, and administration — making a single topology inadequate. A hybrid topology combines the best elements of multiple designs: a fiber-based ring or mesh at the campus backbone level for redundancy, a star topology within each building for manageability, and bus or

tree structures within individual labs for cost-efficiency. At the campus core, high-speed fiber optic cables (single-mode, 10 Gbps or 40 Gbps) interconnect core routers and multilayer switches using a ring or partial mesh arrangement, ensuring that no single link failure isolates an entire building. Each building has its own distribution switch connected to the campus core via dual fiber uplinks. Within each building, a traditional star topology branches from the distribution switch to floor-level access switches, which then connect individual devices via Cat6 cables. This hierarchical three-layer model — core, distribution, and access — is the industry standard for enterprise and campus networks because adding a new building simply means installing a new distribution switch and running fiber to the core, with no disruption to existing infrastructure. Wireless access points throughout the campus connect via PoE (Power over Ethernet) to access switches, eliminating the need for separate power cabling. The hybrid design thus delivers redundancy at the backbone, simplicity at the building level, and scalability campus-wide.

5. Designing a network for a company with multiple departments involves creating a structured and scalable architecture using switches and routers to ensure efficient communication and security. Each department (such as HR, Finance, Sales, and IT) is connected through dedicated Layer 2 switches that form the access layer, allowing devices like computers and printers to communicate within the same department. These switches are then

connected to a central Layer 3 switch or router, which acts as the core layer and enables communication between different departments through inter-VLAN routing. Virtual LANs (VLANs) are configured to logically separate each department's network, improving security and reducing broadcast traffic. A router is also used to connect the internal network to external networks such as the internet, often with firewall functionality for security. Redundant links and backup paths can be included to ensure reliability, while proper IP addressing and subnetting are applied for efficient network management. This hierarchical design improves performance, enhances security, and allows easy expansion as the company grows.

	A	B	C	D	E	F	G	H	
Q ₁	2	3	3	2	2	2	1	1	16
Q ₂	3	3	4	2	2	4	1	0.5	16.5
Q ₃	2	4	4	2	1	1	1	0.5	15.5
Q ₄	2	2	3	1	1	2	0.5	1	12.5
Q ₅	1	3	2	1	2	2	0.5	1	12.5

13

RUBRICS FOR CALCULATION

Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
A Problem Analysis	15%	Clearly identifies and deeply analyzes the industry problem with all factors 3	Good understanding with minor gaps	Basic understanding; some key aspects missing	Poor or incorrect understanding of the problem
B Solution Design	20%	Innovative, practical, and industry-relevant solution 4	Effective solution with minor limitations	Partially suitable solution	Inappropriate or unclear solution
C Technical Implementation	20%	Fully functional, accurate, and well-executed implementation 4	Mostly functional with minor errors	Partially implemented solution	Incomplete or non-functional implementation
D Use of Tools & Technologies	10%	Appropriate and advanced use of relevant tools and technologies 2	Correct use of tools	Limited or basic use of tools	Incorrect or no use of tools
E Problem-Solving Approach	10%	Logical, systematic, and well-structured approach 2	Mostly logical approach	Somewhat disorganized approach	No clear approach
F Innovation & Creativity	10%	Highly creative and original solution 2	Some creativity	Limited originality	No creativity
G Testing & Validation	5%	Thorough testing with real-world scenarios 1	Adequate testing	Minimal testing	No testing
H Documentation & Presentation	5%	Clear, professional, and well-documented 1	Good documentation	Basic documentation	Poor or no documentation
I Teamwork / Collaboration	5%	Excellent coordination and contribution (if team-based) 1	Good collaboration	Limited participation	No collaboration or contribution

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Annexure A

Industry Problem-Solving Task

1. A star topology is ideal for a 20-person startup because all devices connect to a central switch, making the network easy to manage and troubleshoot. At the core sits a Layer 2/Layer 3 managed switch (such as a Cisco Catalyst or Netgear ProSafe) with at least 24 ports to accommodate all workstations, printers, and VoIP phones with room to grow. Each device connects to the switch using Cat6 UTP cables, which support Gigabit speeds up to 100 meters — more than sufficient for any typical office floor plan. The switch uplinks to a router/firewall (such as a Cisco RV series or pfSense appliance) that handles internet access, DHCP, and network address translation. A Wi-Fi access point connected to the switch extends wireless coverage for laptops and mobile devices. The key advantage of this design is fault isolation: if one workstation's cable fails, only that single node goes

offline, leaving all other 19 employees unaffected. Centralised cable management also simplifies future moves, adds, and changes.

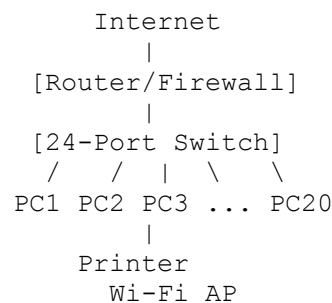
. Network Design for a 20-Person Startup – Star Topology

A **star topology** is suitable for a 20-person startup because all devices are connected to a central switch. This makes the network easy to manage, monitor, and troubleshoot.

Required Components

- 1 Managed Layer 2/Layer 3 Switch with at least 24 ports
- Router/Firewall for internet connectivity and security
- Wi-Fi Access Point
- Cat6 UTP cables
- 20 Computers/Laptops
- Printers and VoIP phones if required

Network Diagram



Working

Every computer is connected directly to the central switch using Cat6 cables. The switch connects to the router/firewall, which provides internet access, DHCP, and network security. The Wi-Fi access point provides wireless connectivity.

Advantages

- Easy to install and manage
- Easy to identify network problems
- Failure of one cable affects only one device
- Easy to add new computers
- Supports future expansion

Conclusion: Star topology is a simple, reliable, and scalable solution for a small startup network.

2. A bank demands near-zero downtime, making a full or partial mesh topology the right choice. In a full mesh, every core network device — routers, core switches, and servers — connects directly to every other device, so no single link or node failure can bring down the network. For a bank, a partial mesh is typically deployed at the core layer (connecting all core switches and firewalls to each other) while the distribution and access layers use redundant uplinks to at least two core switches. Dual enterprise-grade routers (such as Cisco ASR or Juniper MX series) provide BGP-based redundant internet connectivity. All links between core devices use fiber optic cables — either single-mode for long distances between buildings or multi-mode OM4 within the data center — because fiber is immune to electromagnetic interference, supports very high bandwidth, and is essential for maintaining low-latency financial transactions. Redundant power supplies, UPS systems, and RAID storage arrays further complement the mesh design. The justification is straightforward: when a link fails in a mesh, traffic automatically reroutes through alternative paths in milliseconds using protocols like OSPF or BGP, satisfying regulatory requirements for uptime and data integrity.

Network Design for a Bank – Mesh Topology

A bank requires **high availability and near-zero downtime**. Therefore, a **full mesh or partial mesh topology** is the most suitable design.

Required Components

- Redundant Core Switches
- Enterprise Routers
- Firewalls
- Servers
- Fiber Optic Cables

- UPS and Backup Power Supply
- RAID Storage Systems

Network Diagram

[Router 1] ----- [Router 2]

| \ / |

| \ / |

| \ / |

| \ / |

[Core Switch 1]---[Core Switch 2]

| \ / |

| \ / |

| \ / |

[Server 1]----[Server 2]

Working

In a mesh network, important network devices have multiple connections. If one link fails, traffic can automatically use another available path.

Fiber optic cables are used between core devices because they provide high bandwidth, low latency, and protection from electromagnetic interference. Routing protocols such as **OSPF** or **BGP** can help traffic move through alternative paths when failures occur.

Advantages

- Very high reliability
- No single point of failure
- Multiple backup paths
- Fast recovery during link failure

- Suitable for critical banking services

Conclusion: A partial or full mesh topology provides the redundancy and reliability required for banking networks.

3. For a small classroom lab with a tight budget, bus topology offers the lowest hardware cost because all computers share a single common communication line — the bus — eliminating the need for a central switch or hub. A single coaxial cable (RG-58, 50-ohm) or, in a modern variation, a single Cat5e/Cat6 cable daisy-chained through an inexpensive unmanaged hub runs along the length of the room. Each computer connects to the backbone using a T-connector and a BNC connector in a classic coaxial setup, with terminators (50-ohm) placed at both ends of the cable to absorb signals and prevent reflection. For a more contemporary low-cost implementation, a single inexpensive 8- or 16-port unmanaged switch replaces the terminators and uses Cat5e UTP patch cables to each workstation — achieving bus-like simplicity at minimal cost. The primary limitation is that the network is susceptible to collisions and a single cable break disrupts all connected machines, but for a supervised classroom environment with light usage (internet browsing, file sharing), this trade-off is entirely acceptable. No dedicated server is required; a teacher's PC can share resources using built-in Windows/Linux file sharing.

Network Design for a Small Classroom Lab – Bus Topology

For a small classroom with a limited budget, a **bus topology** can be used because it requires less networking hardware.

Required Components

- Computers
- RG-58 Coaxial Cable for classic bus topology
- BNC Connectors
- T-Connectors
- 50-ohm Terminators at both ends

A modern low-cost alternative can use:

- Cat5e/Cat6 cables
- An inexpensive unmanaged switch

Network Diagram

```
[Terminator]-----PC1-----PC2-----PC3-----PC4-----PC5-----[Terminator]
                        Shared Communication Cable
```

Working

All computers share a single communication backbone. Data travels through the common cable, allowing connected devices to communicate.

A teacher's computer can also be used for file sharing and other shared resources without requiring a dedicated server.

Advantages

- Low installation cost
- Requires less hardware
- Simple for small networks
- Suitable for light usage

Disadvantages

- A backbone cable failure can affect the entire network
- More chance of collisions
- Difficult to troubleshoot in larger networks

Conclusion: Bus topology is a low-cost solution for a small classroom lab with limited devices and light network usage.

4. A large university campus serves thousands of users across multiple buildings — lecture halls, labs, libraries, dormitories, and administration — making a single topology inadequate. A hybrid topology combines the best elements of multiple designs: a fiber-based ring or mesh at the campus backbone level for redundancy, a star topology within each building for manageability, and bus or tree structures within individual labs for cost-efficiency. At the campus core, high-speed fiber optic cables (single-mode, 10 Gbps or 40 Gbps) interconnect core routers and multilayer switches using a ring or partial mesh arrangement, ensuring that no single link failure isolates an entire building. Each building has its own distribution switch connected to the campus core via dual fiber uplinks. Within each building, a traditional star topology branches from the distribution switch to floor-level access switches, which then connect individual devices via Cat6 cables. This hierarchical three-layer model — core, distribution, and access — is the industry standard for enterprise and campus networks because adding a new building simply means installing a new distribution switch and running fiber to the core, with no disruption to existing infrastructure. Wireless access points throughout the campus connect via PoE (Power over Ethernet) to access switches, eliminating the need for separate power cabling. The hybrid design thus delivers redundancy at the backbone, simplicity at the building level, and scalability campus-wide.

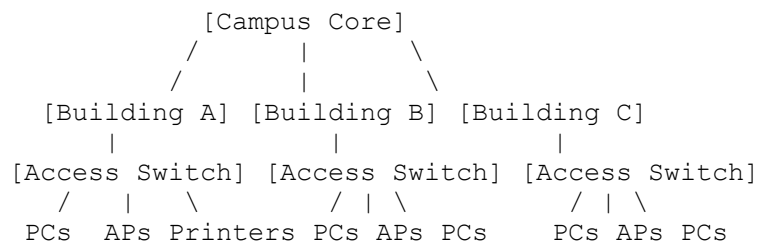
Network Design for a Large University Campus – Hybrid Topology

A large university campus contains multiple buildings, laboratories, libraries, hostels, and administrative departments. A **hybrid topology** is suitable because it combines multiple topologies.

Proposed Design

- **Mesh or Ring Topology** at the campus backbone
- **Star Topology** inside each building
- **Bus or Tree Structure** for individual labs where appropriate
- **Fiber Optic Cables** between buildings
- **Cat6 Cables** for connecting individual devices
- **Wireless Access Points** connected through PoE

Network Diagram



Working

The campus core connects different buildings using high-speed fiber optic cables. Each building has a distribution switch with redundant fiber uplinks. Inside each building, access switches connect computers, printers, and other devices using a star topology.

Wireless access points can use **Power over Ethernet (PoE)**, which provides both network connectivity and power through the Ethernet cable.

Advantages

- Highly scalable
- Reliable backbone connectivity
- Easy to manage individual buildings
- Supports thousands of users
- Easy to add new buildings
- Provides redundancy and fault tolerance

Conclusion: A hybrid topology provides the best combination of scalability, reliability, and easy management for a large university campus.

- 5 Designing a network for a company with multiple departments involves creating a structured and scalable architecture using switches and routers to ensure efficient communication and security. Each department (such as HR, Finance, Sales, and IT) is connected through dedicated Layer 2 switches that form the access layer, allowing devices like computers and printers to communicate within the same department. These switches are then connected to a central Layer 3 switch or router, which acts as the core layer and enables communication between different departments through inter-VLAN routing. Virtual LANs (VLANs) are configured to logically separate each department's network, improving security and reducing broadcast traffic. A router is also used to connect the internal network to external networks such as the internet, often with firewall functionality for security. Redundant links and backup paths can be included to ensure reliability, while proper IP addressing and subnetting are applied for efficient network management. This hierarchical design improves performance, enhances security, and allows easy expansion as the company grows.

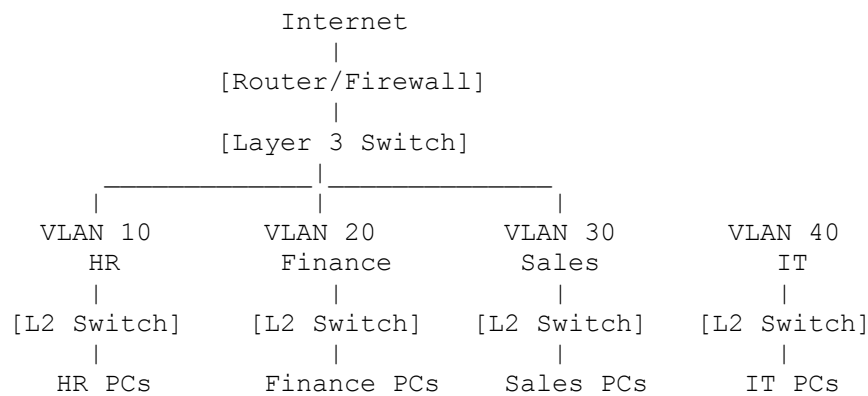
Network Design for a Multi-Department Company

A company with departments such as **HR, Finance, Sales, and IT** requires a structured and secure network design.

Proposed Components

- Layer 2 Access Switches
- Central Layer 3 Switch or Router
- Router/Firewall
- VLAN Configuration
- IP Addressing and Subnetting
- Redundant Links

Network Diagram



Working

Each department is connected to its own Layer 2 switch. VLANs are used to logically separate the departments.

Example:

- **VLAN 10 – HR**
- **VLAN 20 – Finance**
- **VLAN 30 – Sales**
- **VLAN 40 – IT**

A central Layer 3 switch or router performs **inter-VLAN routing**, allowing communication between departments when required. A router/firewall connects the company network to the internet and provides security.

Advantages

- Improved security through VLAN separation
- Reduced broadcast traffic
- Easy network management
- Supports communication between departments
- Easy to expand
- Redundant links improve reliability

Conclusion: A hierarchical network using Layer 2 switches, a Layer 3 core switch, VLANs, routing, and firewall protection provides a secure, scalable, and efficient solution for a multi-department company.

RUBRICS FOR CALCULATION

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Problem Analysis	15%	Clearly identifies and deeply analyzes the industry problem with all factors	Good understanding with minor gaps	Basic understanding; some key aspects missing	Poor or incorrect understanding of the problem
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		well-structured approach			
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